Hot Engines and fire prevention

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I. SIGNIFICANT METHODOLOGIES

Engine fires that start from a hot surface rather than a spark are a specific, well studied problem, and the fix that keeps showing up across the literature is the same id xea applied differently: put a thin ceramic layer, a thermal barrier coating, between the hot metal and the outside world so the surface stays cooler than it would bear. Three papers were picked out of the wider literature review because they each answer a different piece of the same question with real governing equations, real hardware, and real measured numbers. Reference [13] answers the basic version of the question, does a coating actually drop the surface temperature and by how much. Reference [15] asks what happens once the engine stops sitting at one fixed condition and starts driving through a real duty cycle. Reference [16] moves off the piston entirely and asks whether the coating's own manufactured thickness and roughness change how well it performs. Read together, these three papers build a full picture of the problem, and Section IV will show where that picture still has a gap worth filling.

A. Thermal Optimization of Coating Thickness and Material on a Diesel Piston [13].

Reference [13] is the simplest starting point because it isolates the core physical idea before adding any complications. Liu et al. built a finite element model of a diesel piston, which just means the piston shape gets chopped into thousands of small, connected pieces on a computer, and the software solves how heat moves through each piece one at a time. They ran two versions side by side, a bare aluminum piston and the same piston with a thermal barrier coating on top, then compared how hot the surface of each one got under the same combustion conditions.

The physics behind why a coating works at all comes straight from Fourier's law of conduction. Integrating the one-dimensional conduction equation across a plane wall of thickness gives the steady state rate of heat conduction, following Cengel and Ghajar's notation as

$$\dot{Q}_{Cond,\,wall} = kA\frac{T_1 - T_2}{L}\ (W)$$

Rearranging this into the same form as Ohm's law gives the thermal resistance concept

$$\dot{Q}_{cond,\,wall} = \frac{T_1 - T_2}{R_{wall}}$$

where the conduction resistance of the wall is

$$R_{wall} = \frac{L}{kA}\ (K/W)$$

Here is the thermal conductivity of the material, A is the surface area, and L is the thickness. ceramic coating has a much lower than aluminum does, so for that layer is larger even at a small thickness. Stack the coating directly on top of the metal and the two layers behave like two resistors in series, exactly the way Cengel and Ghajar set up a two-layer composite wall with convection on both sides, giving:

$$R_{total} = R_{conv,\,1} + R_{wall,\,1} + R_{conv,\,2} + R_{wall,\,2} = \frac{1}{h_1A} + \frac{L_1}{k_1A} + \frac{L_2}{k_2A} + \frac{1}{h_2A}$$

Where EQ is the convection resistance at each surface, h is the convection coefficient, and the subscripts 1 and 2 mark the coating layer and the base metal layer respectively. The rate of heat transfer through the whole stack then follows the same series relationship:

$$\dot{Q} = \frac{T_{\infty,1} - T_{\infty,2}}{R_{total}}$$

Make the coating thicker, or pick a coating material with a lower k , and R_{total} goes up, which means less heat gets through to raise the surface temperature for the same combustion heat coming from underneath. This textbook relationship is the seed idea behind every paper in this section, since all three are just asking how well this resistance stack holds under different real-world conditions.

Liu et al. did not just pick an arbitrary thickness and material and call it done. They ran a multi objective optimization, testing many combinations of coating thickness and material properties and picking the mix that gave the best tradeoff between two competing goals, maximum temperature reduction against minimum added mechanical stress, since a thicker coating raises and insulates better, but also adds more thermal stress to the piston as it heats and cools. The setup was checked against an actual four-cylinder diesel engine test rig, shown in Figure 4 of the literature review, complete with a common rail fuel injector, cylinder pressure sensor, thermocouple sensors, and a combustion analyzer, so the result was not just a simulation guess sitting on a screen. The measured outcome was a peak surface temperature drop of on the coated piston compared to the bare one, about a 15 percent cut, and the paper notes that thicker ceramic kept giving better insulation, though it also kept adding more thermal stress the thicker it got.

For a project about hot surface ignition, is not a small number. Reference [1] in our literature review found lubricating oil could ignite starting around, so a real 55 degree drop on the piston surface could be the difference between a surface sitting comfortably below that threshold and one sitting dangerously close to it. The clear limitation here, which the paper itself does not dwell on, but which matters for our project, is that this whole test happened at one fixed engine operating point. in the equation above is treated as a constant, but and on the combustion and coolant sides both depend on flow conditions that change constantly once the engine is actually running. The paper tells you the coating works at that one condition, but it does not tell you if the same margin holds once the engine is being driven around instead of sitting at a steady state.

B. Coating Performance Across a Full Drive Cycle [15].

Reference [15] picks up exactly where [13] leaves a gap. Koutsakis and Ghandhi pointed out that a real engine practically never sits at one fixed speed and loads the way a lab test does. It idles, then accelerates, then cruises, then someone floors the throttle, all within a few minutes, and each of those states changes how much heat is coming off combustion and how fast that heat needs to move through the coating. So instead of treating R_{total} as fixed the way [13] effectively does, they built a thermo mechanical model that lets the convection coefficients, and therefore $R_{conv,1}$ and $R_{conv,2}$, update continuously as the engine's operating condition changes throughout a simulated drive cycle, rather than being locked to one steady state value.

The practical difference between the two approaches is worth spelling out. In [13], the model solves the resistance network once and reports one answer. In [15], the same network gets re-solved over and over at each moment of the drive cycle, so the output is not a single surface temperature but a temperature history, of a curve that rises and falls as the simulated driver speeds up and slows down. Their model also tracked the mechanical side at the same time, following how the stress inside the coating shifted as the temperature swings pulled the ceramic and the metal underneath in different directions, which is the same thermal stress tradeoff [13] flagged, now playing out dynamically instead of at one snapshot.

What they found matters directly for this project. A coating that looks perfectly safe while cruising, meaning R_{total} comfortably keeps the surface below the autoignition margin, can behave completely differently the moment the engine is put under hard acceleration, because the heat coming in from combustion spikes faster than the coating's fixed thermal resistance can spread it out. In other words, a single ignition margin number, like the $55.54°C$ from [13], only describes one instant on the resistance network, one specific h_1 and h_2. It does not guarantee the same margin holds five seconds later when the driver hits the gas and the convection coefficients on both sides of the wall shift. The paper's main limitation, much like [13], is that it still treats the coating itself as a fixed, undamaged layer with a constant k and L throughout the whole simulated drive, which is exactly the assumption [16] goes on to question.

C. Coating Thickness and Surface Roughness on Turbine Blade Cooling [16].

Reference [16] shifts the question one more time. Du et al. moved off the piston and onto a turbine blade, and instead of asking whether a coating helps or how it performs across a duty cycle, they asked whether the coating's own manufactured geometry, its thickness and its surface roughness, changes the cooling performance you actually get. Surface roughness is not something that shows up explicitly in the basic formula, but it does change the effective convection coefficient at the coating's outer surface. The air or gas moving past any surface forms a thin boundary layer, a slow moving film of fluid hugging the wall, and that film is itself a resistance that heat has to cross before the moving flow can carry it away. A rougher finish trips and disturbs that film, changing how efficiently heat gets pulled off the surface, so two coatings with the identical and can still end up with different effective values purely because of how their surfaces were finished. Both coating thickness and surface roughness turned out to significantly swing the cooling performance in their tests.

This is the part that connects directly back to a manufacturing quality control angle. A coating spec on paper might call for a fixed L and a smooth finish, which is exactly what assumes when [13] and [15] both plug in a single design thickness. But if the manufacturing process does not hit that exact L or that exact surface finish every time, the effective for a real, physically built part can drift from what the design model promised, even though the geometry looks correct on a drawing. That is the same worry [13] and [15] do not fully address, because both of them modeled a coating built exactly to its intended design rather than asking what happens once real manufacturing variation is added on top.

D. Summerizing the three together

Line these three up and a clear pattern shows up, and it reads almost like the same resistance network being stress tested one assumption at a time. Reference [13] proves the basic physics works and gives a real measured number to design toward but holds every term in constant. Reference [15] releases the convection terms and shows the margin moves once and start changing throughout a full duty cycle instead of sitting fixed. Reference [16] releases the coating geometry itself and shows that even a perfectly designed on paper can underperform once real manufacturing variation changes the actual and surface finish that ends up on the part. Each paper relaxes one assumption the previous one kept, but none of the three relaxes all of them at once. That gap, a coating thermal resistance model that holds up across a full duty cycle and stays reliable even with realistic manufacturing tolerance, is exactly the space our proposed methodology in Section IV is aiming to fill.

Table 1. Comparative Summary of Thermal Barrier Coating Approaches:

Ref	**Approaches**	**Relationships**	**Results**	**Limitation**
[13]	**Finite element model, bare vs coated diesel piston, multi objective thickness and material**	**R_total = R_conv,1 + R_wall,1 + R_wall,2 + R_conv,2, held fixed at one operating point**	**Peak surface temperature reduced 55.54°C (15.08 percent) versus bare aluminum piston**	**Tested at a single fixed operating point, h1 and h2 not varied**

	optimization			
[15]	**Thermomechanical model tracking coating behavior across a full simulated drive cycle**	**Same resistance network, but R_conv,1 and R_conv,2 updated continuously with load and speed**	**Coating performance and ignition margin shift significantly with changing operating conditions**	**Assumes L and k of the coating stay fixed and undamaged throughout**
[16]	**Studied the effect of coating thickness and surface roughness on turbine blade cooling performance**	**R_wall,1 = L/(kA), with surface roughness affecting the effective h at the outer boundary**	**Both thickness and roughness significantly changed cooling performance sensitivity**	**Tested on turbine blades rather than piston or engine bay surfaces, needs translation to fit this project**

II. PROPOSED METHODOLOGY

A. Proposed Computational Maping Approach and Justification

To systematically prevent hot- surface engine fires during the design phase, we propose a multi-dimensional, predictive computational thermal mapping framework. Rather than relying on physical trial-and-error, this approach simulates the coupled effects of conduction, convection, and radiation across high-risk under hood components (such as the exhaust manifold and turbocharger housing) [18]. By mapping these surface temperatures, we can proactively assess the structural margin against the autoignition thresholds of common volatile fluids (such as lubricating motor oil at $311°C$ to $407°C$ [1]

First, the system geometry and material thermal properties are defined, Second, convective and radiative boundary conditions are applied based on realistic engine operating loads. Third, the numerical solver resolves the steady-state and transient heat transfer equations. Fourth, a localized thermal map is generated to isolate components operating at elevated temperatures. Finally, the safety margin is calculated as $T_{marign} = T_{auto} - T_S$
. This systematic sequence is justified because it isolates design defects such as unshielded zones prone to post-shutdown heat soak [18] long before physical prototyping begins.

B. Governing Equation and Analytical Derivations

The physics governing our predictive framework is rooted in the classical conservation of energy. For a composite engine wall coated with a thermal barrier coating (TBC) [13] under operating conditions, the steady-state heat transfer rate per unit area can be derived by modeling the system as a series of thermal resistances:

$$R_{\text{total}} = R_{\text{conv,in}} + R_{\text{cond,substrate}} + R_{\text{cond,TBC}} + R_{\text{comb,out}}$$

Where each individual resistance term is derived from conduction and convection fundamentals:

$$R_{conv,in} = \frac{1}{h_{in}A}$$

$$R_{\text{cond,substrate}} = \frac{L_{\text{substrate}}}{k_{\text{substrate}}A}$$

$$R_{\text{cond,TBC}} = \frac{L_{\text{TBC}}}{k_{\text{TBC}}A}$$

At the outer engine surface exposed to the under-hood compartment, heat is dissipated via parallel pathways of convection to the moving air and radiation to the surrounding firewall or engine housing:

$$q_{\text{total}} = h_{\text{out}}A(T_s - T_\infty) + \epsilon\sigma A(T_s^4 - T_{\text{surr}}^4)$$

To simplify the numerical solution, the radiation term is linearized by defining a radiation heat transfer coefficient (h_{rad}):

$$h_{\text{rad}} = \epsilon\sigma(T_s + T_{\text{surr}})(T_s^2 + T_{\text{surr}}^2)$$

This yields a combined external thermal resistance of:

$$R_{\text{comb,out}} = \frac{1}{(h_{\text{out}} + h_{\text{rad}})A}$$

By coupling these equations, the local outer surface temperature (T_s) of the engine component can be predicted analytically:

$$T_s = T_\infty + \frac{q_{\text{total}}}{(h_{\text{out}} + h_{\text{rad}})A}$$

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This model mathematically demonstrates how increasing the thermal resistance of the barrier layer [13] directly decreases the resulting surface temperature (T_s) , keeping it safely below the fluid's autoignition threshold [1].

C. Governing Equation and Analytical Derivations

To guarantee the mathematical and physical accuracy of our predictive framework, a rigorous verification and validation (V&V) protocol will be implemented.

- **Verification:** Numerical verification will be performed via a grid convergence index (GCI) study. We will run the model on three progressively finer meshes to ensure that the calculated temperature fields are independent of spatial discretization errors.
- **Validation:** The computational model will be validated by direct comparison against empirical benchmark data. Specifically, the predicted outer surface temperatures will be cross-referenced with the experimental under hood soak data from Vdovin and Kadri Sathiyan [18], ensuring that the transient temperature rise observed during post-shutdown heat soak matches physical test-rig measurements.

D. -- Operational Advantages and Research Limitations

Our proposed thermal mapping methodology offers several key benefits, but it is also subject to specific operational constraints.

- **Scenarios Where Effective:** This approach is highly effective during the pre-production design phase for evaluating static, high-load engine states (such as steady highway cruising) [15]. It is also uniquely valuable for analyzing post-shutdown heat soak, where radiative transfer dominates and physical probes are difficult to place [18].
- **Limitations & Failures:** The model will fail or lose accuracy during highly chaotic, transient fluid events such as a pressurized oil leak spraying dynamic, unsteady droplets onto a hot surface [3]. Because the liquid oil film dynamically changes the local convective heat transfer coefficient through boiling and evaporation, the static resistance model cannot accurately predict surface temperatures without real-time multiphase CFD coupling [3, 4].

E. Abbreviations and Acronyms

Thermal Barrier Coating (TBC)

Computational Fluid Dynamics (CFD)

Grid Convergence Index (GCI)

III. Implementation

The methodology proposed in Section II is implemented through two coordinated tracks: a computational track, in which the complete thermal model is built and solved in SolidWorks, and a physical track, a low-cost benchtop hot-surface rig used to validate the simulation before its predictions are trusted. This section describes the software toolchain, the proposed hardware, the solution algorithm, and the proposed experimental setup.

A. Software

All thermal modeling is performed in SolidWorks with the SolidWorks Simulation thermal module. SolidWorks was selected for three reasons. First, it combines CAD and finite element analysis (FEA) in a single environment, so design changes such as an added thermal barrier coating (TBC) layer or a repositioned heat shield can be re-meshed and re-solved without exporting geometry between programs. Second, its thermal solver supports both steady-state and transient studies with conduction, convection, and radiation boundary conditions, which are exactly the three heat transfer modes in the governing equations of Section II. Third, the finite element approach follows the precedent of Liu et al., who used a finite element piston model to optimize TBC designs in diesel engines [13].

Within SolidWorks, material properties (k, ρ, cp) are assigned from the built-in library for aluminum alloy, gray cast iron, and steel; convection coefficients are entered from textbook correlations; and surface emissivities are taken from published tables. Mesh independence is enforced with the grid convergence procedure described in Section II.

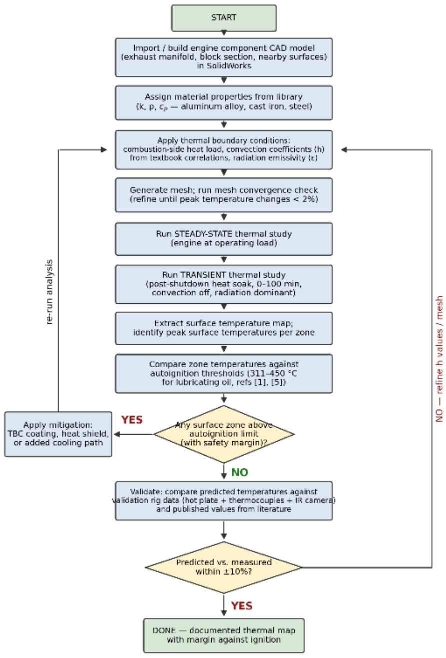
Two supporting programs complete the toolchain. The Arduino IDE (C/C++) is used to write the data acquisition (DAQ) firmware that polls the thermocouple converters at 1 Hz, and Python (pandas and Matplotlib) is used on the logging laptop to time-align thermocouple and infrared (IR) camera data, plot heat-soak curves, and compute the predicted-versus-measured error against the ±10% acceptance criterion. All predicted surface temperatures are ultimately screened against the 311–450 °C autoignition band reported for lubricating oils [1], [5].

B. Hardware

The proposed validation hardware is the benchtop hot-surface rig shown in Fig. 6. It is modeled after the hot-plate ignition experiments used throughout the literature [2], [6] and the simplified test-rig philosophy of Udovin and Kadri Sathiyan, who showed that a reduced-scale rig can reproduce underhood thermal behavior at a fraction of the cost of a full-vehicle test [18]. An electric hot plate rated to approximately 500 °C drives the sample above the full 311–450 °C ignition band of interest [1], [5]. The heated sample plates are 6061 aluminum and gray cast iron, matching the material families assigned in the simulation. Three K-type thermocouples placed at the center, mid-radius, and edge of the sample capture the in-plane temperature gradient; each is read by a MAX31855 cold-junction-compensated converter connected to an Arduino Uno, and the data stream is logged on a laptop. An IR thermal camera aimed at the sample provides a full-field surface temperature map directly comparable to the simulated surface plots; emissivity error is controlled by painting a high-emissivity reference patch on each sample and calibrating the camera against the center thermocouple. No fuel or oil is ever introduced onto the rig—measured temperatures are compared only against published ignition thresholds, which keeps the experiment safe for a standard laboratory setting.

C. Solution Flow Chart

Fig. 5 presents the complete solution algorithm. The engine component CAD model (exhaust manifold, block section, and adjacent shielding surfaces) is built or imported, materials are assigned, and the thermal boundary conditions are applied. The mesh is then refined until the peak predicted temperature changes by less than 2% between refinements. A steady-state study simulates the engine at operating load, followed by a transient study of the 0–100 minute post-shutdown heat soak, during which forced convection is removed and radiation dominates [18]. Peak surface temperatures are extracted per zone and compared against the autoignition thresholds with the safety margin Tmargin = Tauto − Ts [1].



The algorithm contains two feedback loops. In the mitigation loop, any zone exceeding its limit triggers a design change—a TBC layer [14], [15], a heat shield, or an added cooling path [17]—and the analysis is re-run on the modified geometry. In the validation loop, if the predicted temperatures disagree with rig measurements by more than ±10%, the convection coefficients and mesh are refined and the model is re-solved. The procedure terminates only when the design both clears the ignition margin and matches experiment.

Fig. 5. Solution flow chart of the proposed SolidWorks thermal mapping and validation algorithm.

D. Proposed Experimental Setup

Fig. 6 shows the proposed test configuration in side view. The test procedure mirrors the two simulation studies. For the steady-state comparison, the hot plate is stepped through 150, 250, 350, and 450 °C setpoints and held at each until the thermocouple readings drift by less than 1 °C per minute; the settled readings and IR map at each setpoint provide four steady-state validation points. For the transient comparison, the plate is then switched off and the natural cool-down is logged for 100 minutes, reproducing the radiation-dominated heat-soak decay observed in underhood measurements [18]. Finally, the rig itself is replicated as a simple SolidWorks model, so that measured and simulated values are compared on identical geometry—isolating solver accuracy from the geometric uncertainty of the full engine model.

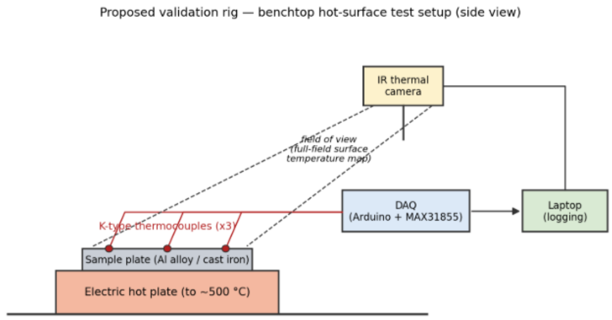


Fig. 6. Proposed benchtop hot-surface validation rig (side view): electric hot plate, instrumented sample plate, K-type thermocouples with Arduino/MAX31855 DAQ, and IR thermal camera providing a full-field surface temperature map.

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